

# Lecture 7: Trigonometric Substitutions

Monday, September 21 • Sections 3.3

• Guided Notes •

**How to Use These Guided Notes:** These guided notes are designed to help you organize the main ideas from each lecture but keep in mind that

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

Trigonometric substitutions are a special type of substitution. In the substitutions we have used so far, we defined the new variable in terms of  $x$ . For example, we used  $u = x + 1$  or  $u = x^2 - 1$ . With trigonometric substitution, we reverse this relationship and define  $x$ , the original variable, in terms of a new variable, usually  $\theta$ , using a trigonometric function  $f$ :  $x = f(\theta)$ .

The trigonometric function  $f$  that we use depends on the integrand. For example, an expression like  $x^2 - 9$  suggests the substitution  $x = 3 \sec(\theta)$ . An expression like  $25 - 2x^2$  suggests  $\sqrt{2}x = 5 \sin(\theta)$ , while  $x^2 + 11$  suggests  $x = \sqrt{11} \tan(\theta)$ . Use the table below to choose a trigonometric substitution based on the expression you see in the integrand. In the table,  $a$  and  $b$  represent positive real numbers.

| When to use it | Trig substitution     | $dx$                                                 |
|----------------|-----------------------|------------------------------------------------------|
| $(ax)^2 - b^2$ | $ax = b \sec(\theta)$ | $dx = \frac{b}{a} \sec(\theta) \tan(\theta) d\theta$ |
| $b^2 - (ax)^2$ | $ax = b \sin(\theta)$ | $dx = \frac{b}{a} \cos(\theta) d\theta$              |
| $(ax)^2 + b^2$ | $ax = b \tan(\theta)$ | $dx = \frac{b}{a} \sec^2(\theta) d\theta$            |

## Example 7.1

Let us write the corresponding trigonometric substitution.

| Integral                                                             | Trigonometric substitution |
|----------------------------------------------------------------------|----------------------------|
| $\int \frac{1}{\sqrt{25+x^2}} dx$                                    |                            |
| $\int \frac{\sqrt{9-x^2}}{x^2} dx$                                   |                            |
| $\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2+9)^{\frac{3}{2}}} dx$ |                            |
| $\int \frac{x^3}{\sqrt{1-x^2}} dx$                                   |                            |
| $\int \frac{1}{x^2\sqrt{x^2+4}} dx$                                  |                            |
| $\int \frac{x}{\sqrt{x^2+4}} dx$                                     |                            |
| $\int \frac{x^2}{\sqrt{x^2-2}} dx$                                   |                            |
| $\int \frac{1}{\sqrt{25x^2+9}} dx$                                   |                            |

**! Keep in mind**

Substitution  $ax = b \tan(\theta)$

**When to use it:**  $b^2 + a^2x^2$ .

**Using a trigonometric identity to rewrite integral:** Since  $\tan^2(\theta) + 1 = \sec^2(\theta)$ , the expression  $b^2 + a^2x^2$  can be rewritten as

$$b^2 + a^2x^2 = b^2 + b^2 \tan^2(\theta) = b^2 (1 + \tan^2(\theta)) = b^2 \sec^2(\theta).$$

**Associated inverse trigonometric function:** Inverse tangent or arc tangent.

1. The substitution defined by  $ax = b \tan(\theta)$  is equivalent to

$$\tan(\theta) = \frac{ax}{b} \quad \text{and} \quad \theta = \arctan\left(\frac{ax}{b}\right).$$

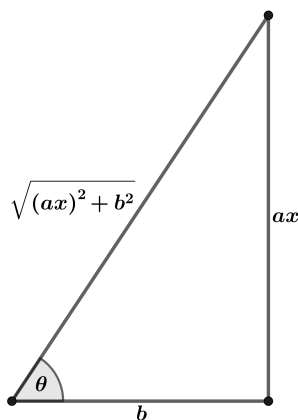
2.  $\theta \in \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[ \Rightarrow \sec(\theta) > 0$ .

3. **Representative triangle:** In a right triangle,  $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$ .

$$\cos(\theta) =$$

$$\sin(\theta) =$$

$$\sec(\theta) =$$



**Example 7.2**

$$\int \frac{1}{x^2 \sqrt{x^2 + 4}} dx$$

**Substitution Properties:**

1.  $x = 2 \tan(\theta)$ .

2.  $\tan(\theta) =$  \_\_\_\_\_  $\theta =$  \_\_\_\_\_

3.  $\theta \in \rule{1.5cm}{0.4pt} \Rightarrow \sec(\theta) > 0.$

**Applying the substitution:**

$$\sqrt{x^2 + 4} =$$

$$x^2 \sqrt{x^2 + 4} =$$

$$dx =$$

**New integral (in terms of the new variable,  $\theta$ ):**

**Evaluating new integral:**

**Rewriting the result in terms of  $x$ :**

**! Keep in mind**

Substitution  $ax = b \sin(\theta)$

**When to use it:**  $b^2 - a^2x^2$ .

**Using a trigonometric identity to rewrite the integral:** Since  $1 - \sin^2(\theta) = \cos^2(\theta)$ , the expression  $b^2 - a^2x^2$  can be rewritten as

$$b^2 - a^2x^2 = b^2 - b^2 \sin^2(\theta) = b^2 (1 - \sin^2(\theta)) = b^2 \cos^2(\theta).$$

**Associated inverse trigonometric function:** Inverse sine or arc sine.

1. The substitution defined by  $ax = b \sin(\theta)$  is equivalent to

$$\sin(\theta) = \frac{ax}{b} \quad \text{and} \quad \theta = \arcsin\left(\frac{ax}{b}\right).$$

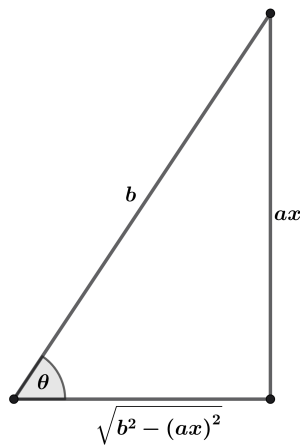
2.  $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \Rightarrow \cos(\theta) \geq 0$ .

3. **Representative triangle:** In a right triangle,  $\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}$ .

$$\cos(\theta) =$$

$$\tan(\theta) =$$

$$\sec(\theta) =$$

**Example 7.3**

$$\int_{-\frac{1}{2}}^{\frac{\sqrt{2}}{2}} \frac{x^3}{\sqrt{1-x^2}} dx$$

**Substitution Properties:**

1.  $x = \sin(\theta)$ .

2.  $\theta =$  \_\_\_\_\_

3.  $\theta \in$  \_\_\_\_\_  $\Rightarrow \cos(\theta) \geq 0$ .

Applying the substitution and writing the new integral (in terms of the new variable,  $\theta$ ):

Evaluating new integral:

Rewriting the result in terms of  $x$ :

**! Keep in mind**

Substitution  $ax = b \sec(\theta)$

**When to use it:** \_\_\_\_\_

**Using a trigonometric identity to rewrite the integral:** Since  $\sec^2(\theta) - 1 = \tan^2(\theta)$ , the expression  $a^2x^2 - b^2$  can be rewritten as

$$b^2 \sec^2(\theta) - b^2 = b^2 (\sec^2(\theta) - 1) = b^2 \tan^2(\theta).$$

**Associated inverse trigonometric function:** Inverse secant or arc secant.

1. The substitution defined by  $ax = b \sec(\theta)$  is equivalent to

$$\sec(\theta) = \frac{ax}{b} \quad \text{and} \quad \theta = \operatorname{arcsec}\left(\frac{ax}{b}\right).$$

2.  $\theta \in [0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi] \Rightarrow \tan(\theta)$  may be positive or negative.

If  $x \geq \frac{b}{a}$ , then  $\theta \in [0, \frac{\pi}{2})$  and  $\tan(\theta) \geq 0$ .

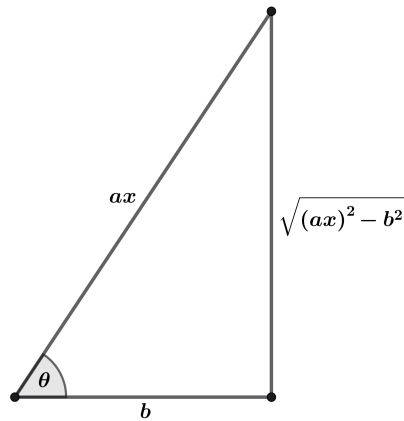
If  $x \leq -\frac{b}{a}$ , then  $\theta \in (\frac{\pi}{2}, \pi]$  and  $\tan(\theta) \leq 0$ .

3. **Representative triangle:** In a right triangle,  $\sec(\theta) = \frac{1}{\cos(\theta)} = \frac{\text{hypotenuse}}{\text{adjacent}}$ .

$$\sin(\theta) =$$

$$\tan(\theta) =$$

$$\cos(\theta) =$$



**One word of precaution:** A reference triangle uses positive side lengths. While the values we obtained before are also true when  $x$  is negative, this is not the case anymore. If  $x < 0$ ,  $\tan(\theta) = \tan\left(\operatorname{arcsec}\left(\frac{ax}{b}\right)\right)$  is a negative number and therefore

$$\tan(\theta) = -\frac{\sqrt{a^2x^2 - b^2}}{b}.$$

**Example 7.4**

Determine  $\int \frac{x^2}{\sqrt{x^2 - 2}} dx$  with  $x > \sqrt{2}$ .

**Substitution Properties:**

1.  $x = \sqrt{2} \sec(\theta)$ .
2.  $\theta =$  \_\_\_\_\_
3. Because  $x > \sqrt{2}$ ,  $\theta \in$  \_\_\_\_\_  $\Rightarrow \tan(\theta) \geq 0$ .

**Applying the substitution and writing the new integral (in terms of the new variable,  $\theta$ ):**

**Evaluating new integral:**

**Rewriting the result in terms of  $x$ :**

**Example 7.5**

Evaluate

$$\int \frac{\sqrt{9-x^2}}{x^2} dx.$$

Let us practice these substitutions avoiding the following common mistakes

**! Common Mistakes**

1. Forgetting to replace  $dx$ .
2. Choosing a substitution that does not produce a useful Pythagorean identity.
3. Mixing expressions involving  $x$  and  $\theta$  in the same integral.
4. Ignoring the interval when determining the signs of the trigonometric functions, especially when using a secant substitution.
5. Using the reference triangle to determine a sign instead of using the interval for  $\theta$ .
6. Forgetting to rewrite the final answer in terms of the original variable.
7. Using the original  $x$ -bounds in an integral written in terms of  $\theta$ .

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**Practice Problems**

After applying integration by parts to the first three problems, you may use a trigonometric substitution to solve the resulting integral, but feel free to use another appropriate method.

1. Determine

$$(a) \int x \sin^{-1}(x) \, dx.$$

$$(b) \int \frac{x \ln(x)}{\sqrt{x^2 - 1}} \, dx.$$

$$(c) \int \ln(1 + x^2) \, dx.$$

$$(d) \int_{-5}^5 \sqrt{25 - x^2} \, dx.$$

$$(e) \int \sqrt{1 - 9x^2} \, dx.$$

$$(f) \int \frac{8}{x^2 \sqrt{4 - x^2}} \, dx.$$

$$(g) \int_0^{\frac{3}{2}} \frac{8}{(x^2 + 1)^2} \, dx.$$

$$(h) \int \frac{2}{x^3 \sqrt{x^2 - 1}} \, dx, \quad x > 1.$$

$$(i) \int \frac{\sqrt{y^2 - 25}}{y^3} \, dy, \quad y > 5.$$

$$(j) \int \frac{5}{\sqrt{25x^2 - 9}} \, dx, \quad x > \frac{3}{5}.$$

$$(k) \int \frac{1}{x^2 \sqrt{x^2 - 4}} \, dx.$$

2. Find the area of the ellipse defined by  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ .

**By the end of today you will be able to:**

- Choose an appropriate sine, tangent, or secant substitution based on the form of the integrand.
- Rewrite an integral completely in the new variable.
- Use a reference triangle to return to the original variable.
- Distinguish trig substitution from ordinary substitution and trig integrals.